BOSTON UNIVERSITY

EC 551 – Advanced Digital Design with Verilog and FPGAs

Lecture 9
Behavioral Synthesis

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Additional material courtesy of Alexander Taubin, EC551 Fall 2009

Alternative State Machine Representations

Why State Diagrams (or State-transition graphs (STG)) Are Not Enough

- Not flexible enough for describing very complex finite state machines
- Not suitable for gradual refinement of finite state machine
- Do not obviously describe an *algorithm:* that is, well specified sequence of actions based on input data
  - algorithm = sequencing + data manipulation
  - separation of control and data
- *Gradual shift towards program-like representations:*
  - Algorithmic State Machine (ASM) Chart
  - Hardware Description Languages (e.g., Verilog)

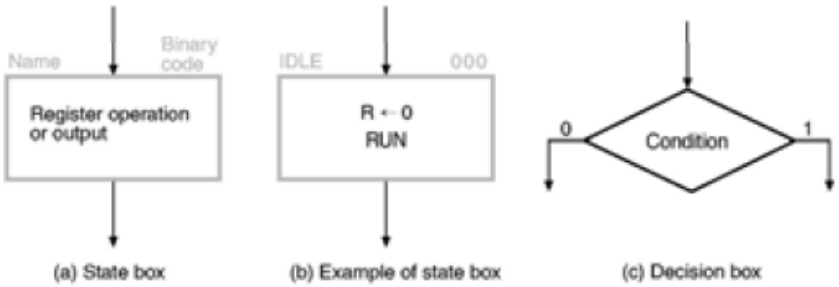
Algorithmic State Machine (ASM) Chart

- Similar to software flowcharts but display the time sequence of computation
- ASM charts reveal the sequential steps of a machine's activity under the influence of machine's inputs
- Focus on machine's activity (unfolds during the operations), rather than contents of registers (data that are produced)

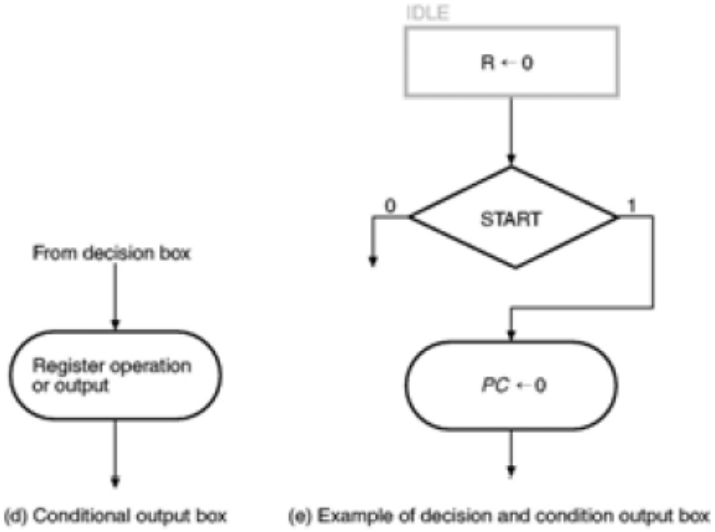
Example: instead of describing a 16-bit counter by its content, we can view it as a datapath unit and describe its activity (e.g., counting, waiting, etc.).

- ASM chart elements
  - state box
  - decision (condition) box
  - conditional output or register operation box
- Clock governs transitions between states
- ASM charts represent Mealy and Moore machines
- For Moore-type the outputs are usually listed inside a state box
- Conditional outputs (Mealy outputs) are placed in a conditional box

ASM Chart Fundamentals



(a) State box (b) Example of state box (c) Decision box



(d) Conditional output box (e) Example of decision and condition output box

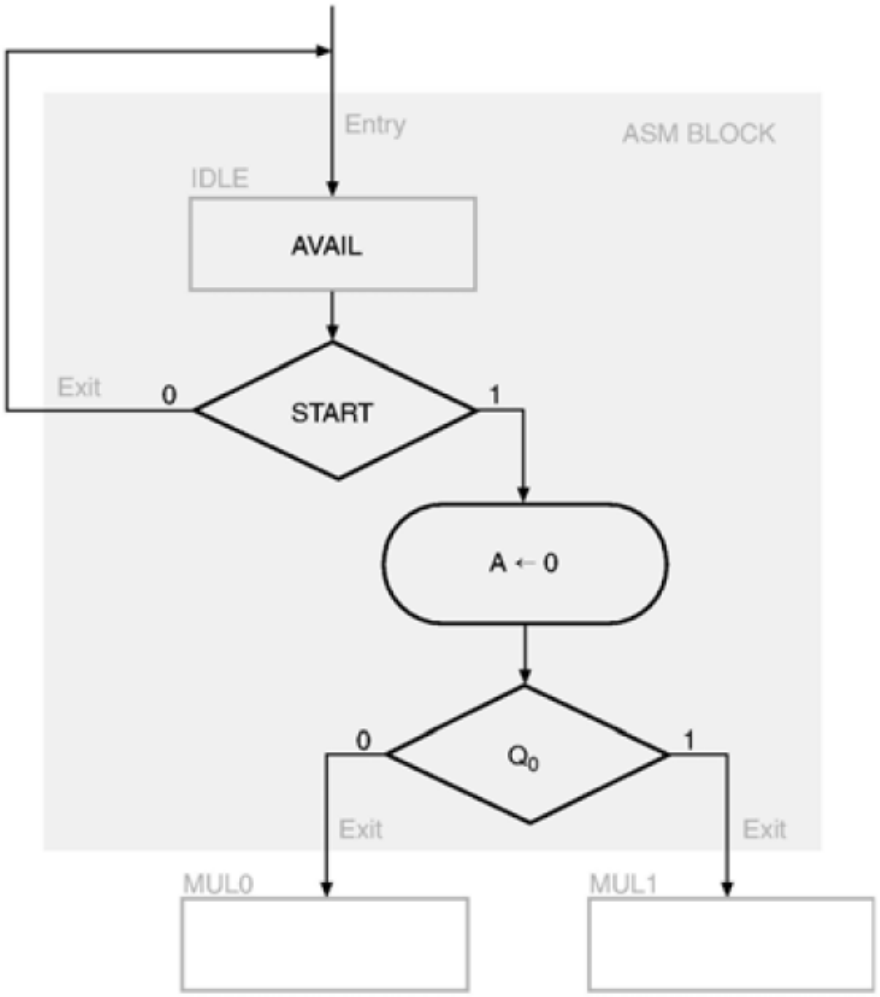
Hardware algorithm: A sequence of procedural steps that perform a data processing task

- Flowchart - specifies a sequence of procedural steps and decision paths for the algorithm
- Finite State Machine = a sequential digital circuit
- Algorithm State Machine = use of a flowchart to define digital hardware algorithms
  - describes a sequence of events and decision paths
  - describes the timing relationship between states

ASM Chart has arrows and three types of boxes :

- state box -- rectangle
  - Contains register transfer operations and/or outputs.
  - Activated while control unit is in that state
- decision box – diamond shaped
  - Single Boolean expression involving a decision with two paths corresponding to T/F evaluation.
- conditional output box -- oval shaped
  - Specifies action to be taken if conditions from up-stream decision box(es) is (are) met
  - Entry must come from a decision box

ASM Block



ASM Block ➔

- one state box and all the decision and conditional output boxes connected between the state box's exit and entry paths to the same or to other state boxes
- each state box is equivalent to a node in the state diagram for the sequential circuit part of the Control Unit
- timing: assume all Flip-Flop's used are positive-edge triggered

ASM chart vs. State Diagram ➔

- each state box is a node in the state diagram
- decision box gives input value on the lines connect nodes in state diagram
- outputs of state box are those of the state node in state diagram
- outputs of conditional box are those of input values on the lines connecting states in state diagram

Algorithmic State Machine (ASM) Notation

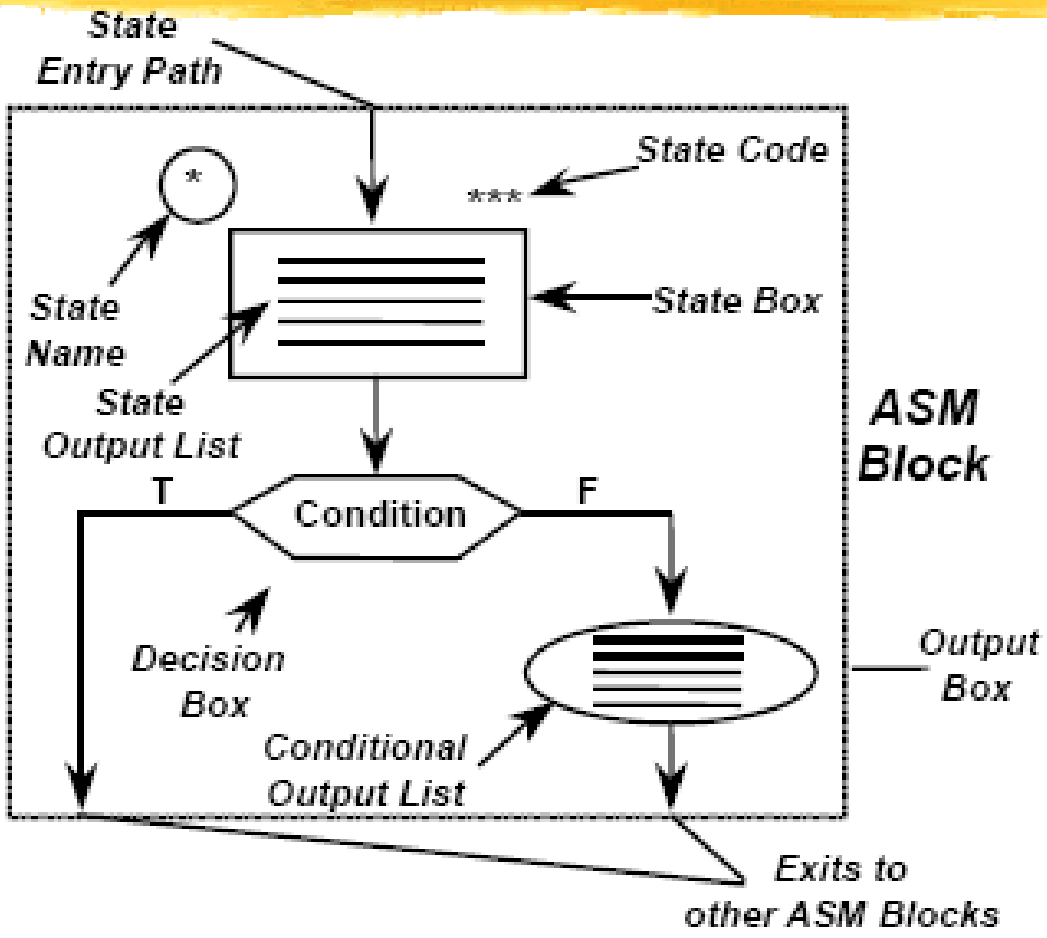
State Machine in one state block per state time

The state box represents the state of machine between synchronizing clock events

Single Entry Point

Unambiguous Exit Path for each combination of inputs (value of variable in decision box)

Outputs asserted high (.H) or low (.L); Immediate (I) or delayed till next clock



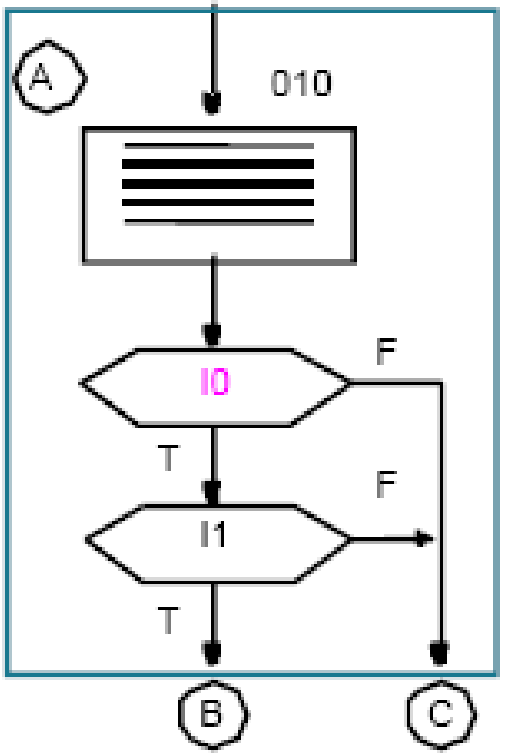
ASM Notation

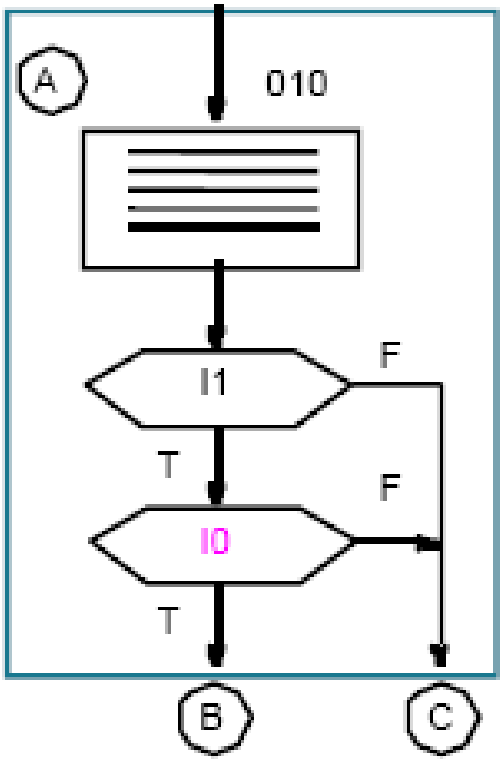
Decision Boxes:

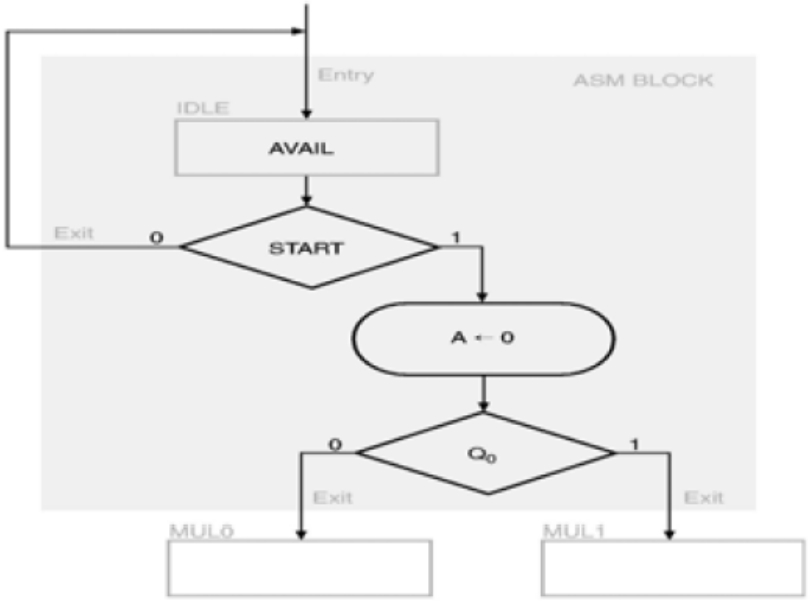
Ordering has no effect on final outcome

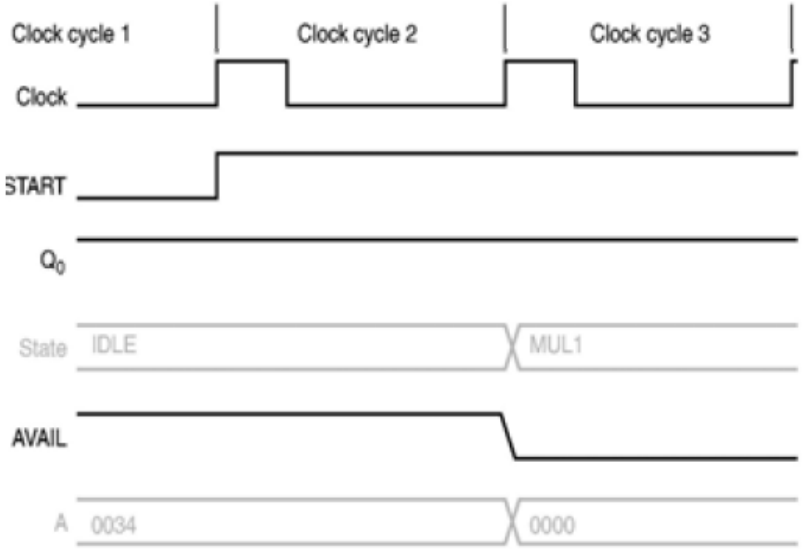
Equivalent ASM charts:

A exits to B on (I0 • I1) else exit to C









ASM Timing Behavior

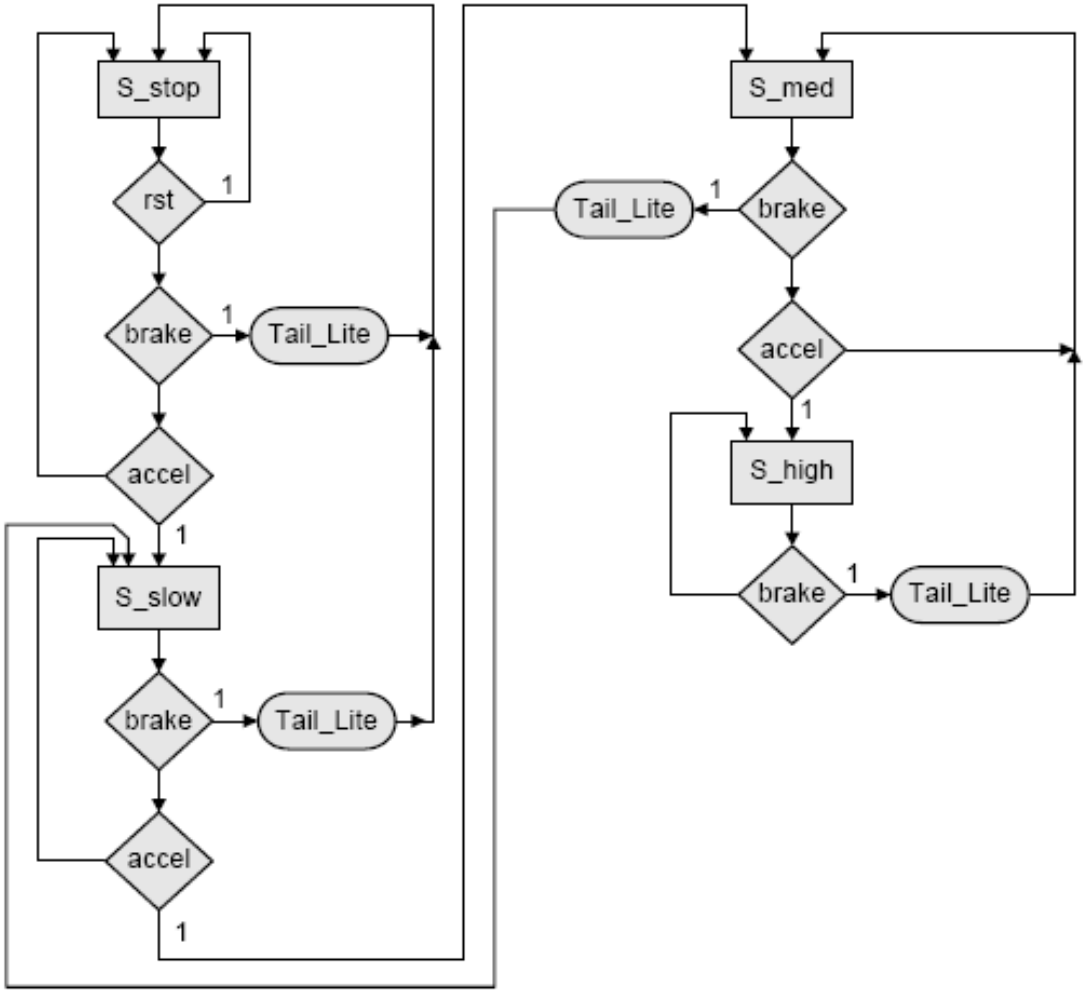
Initially (clock cycle 1)

- In the IDLE state
- Control variable START is 0 (false)
- At every clock positive edge, AVAIL is set to 1

When variable START becomes 1 (true), the following happen at the next positive-going clock edge:

- Register A is cleared (set to 0)
- We leave state IDLE and go to either state MUL0 or state MUL1 depending on the value of Qo
  - since Qo is high, we enter state MUL1
- AVAIL goes to 0 (becomes false) when we exit state IDLE

Example: Vehicle Speed and Tail Light Controller



Parity Checker Example

A string of bits has "even parity" if the number of 1's in the string is even.

- Design a circuit that
  - accepts a bit-serial stream of bits (In) and
  - outputs a 0 if the parity thus far is even and outputs a 1 if odd (out):



Example: in: start 0 1 0 1 1 1 0 0

even even odd odd even odd even even even

bit flow

- Can you guess a circuit that performs this function?

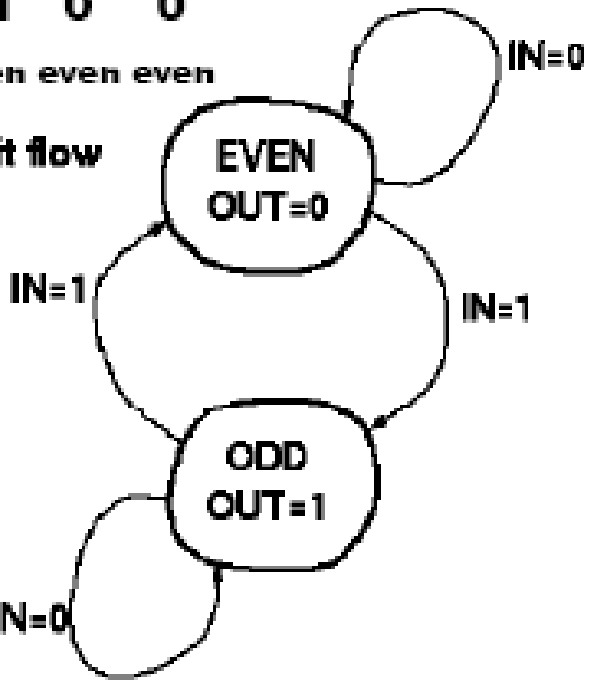
Formal Design Process . Moore machine



Example: in: start 0 1 0 1 1 1 0 0

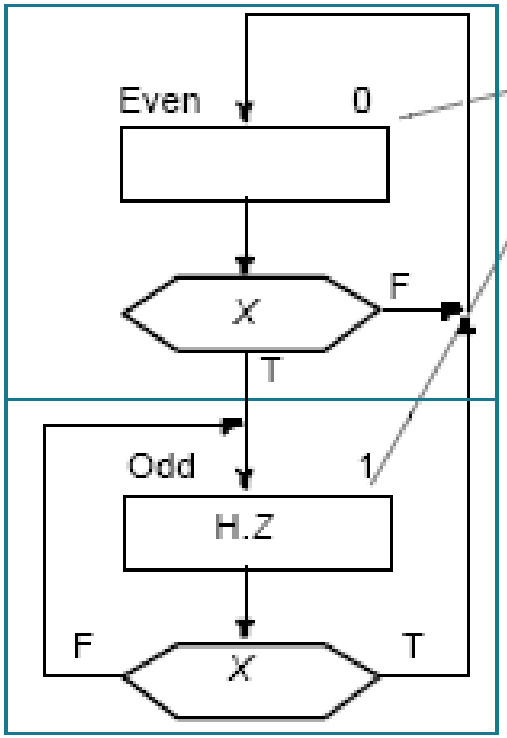
even even odd odd even odd even even even

bit flow



- "State Transition Diagram"
  - circuit is in one of two states.
  - transition on each cycle with each new input over exactly one arc (edge).
  - Output depends on which state the circuit is in.

Example: Parity Checker



Trace paths to derive state transition tables

Input X, Output Z

Nothing in output list implies Z not asserted

Z asserted in State Odd

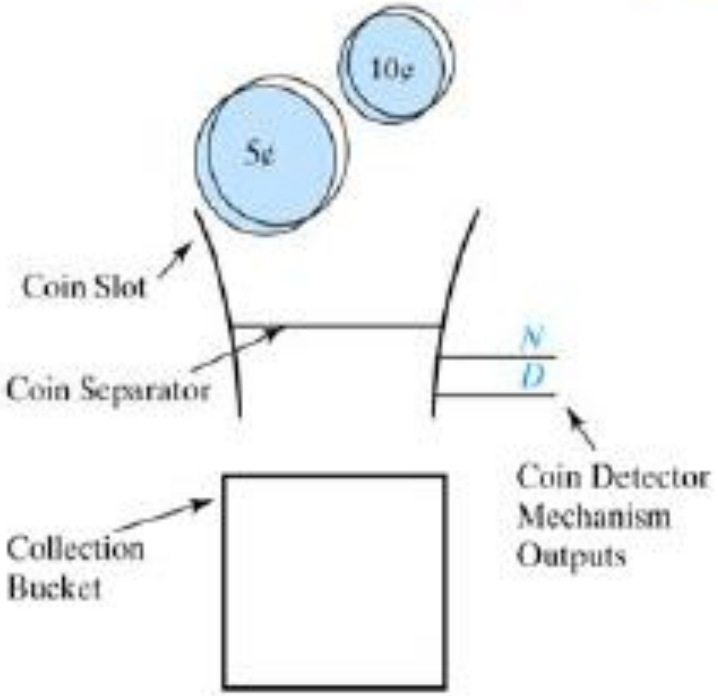
Symbolic State Table:

Input	*Present State*	*Next State*	*Output*
F	Even	Even	—
T	Even	Odd	—
F	Odd	Odd	A
T	Odd	Even	A

Encoded State Table:

Input	*Present State*	*Next State*	*Output*
0	0	0	0
1	0	1	0
0	1	1	1
1	1	0	1

Coin sensor and release mechanism





Basic Design Approach

Example: Vending Machine FSM

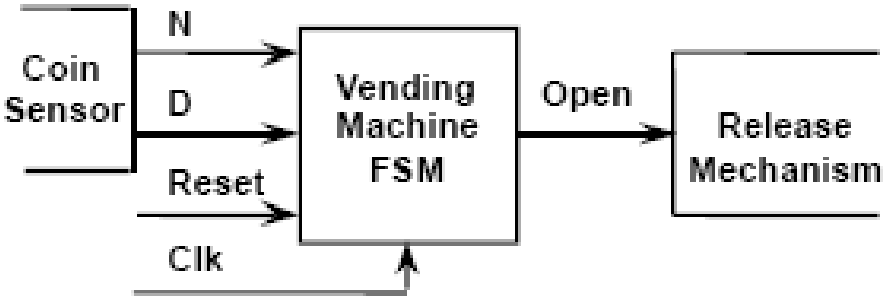
General Machine Concept:

deliver a candy bar after 15 cents deposited

single coin slot for dimes, nickels

no change

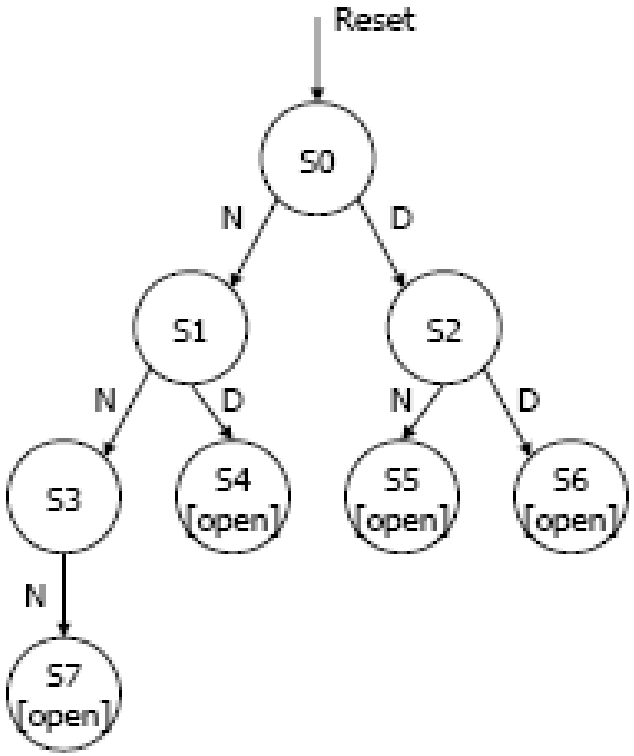
Step 1. *Understand the problem:*



Block Diagram

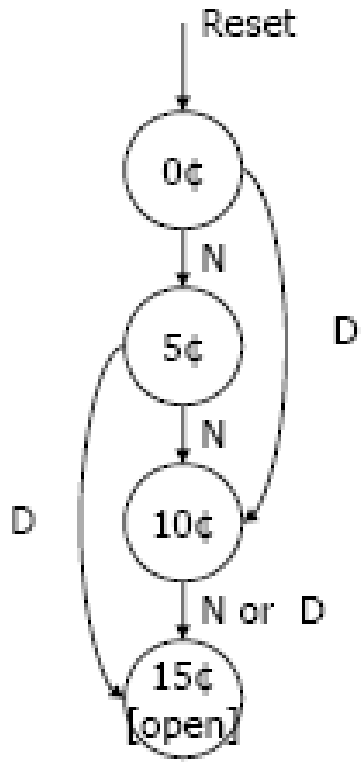
Example: vending machine (cont'd)

- Suitable abstract representation
  - tabulate typical input sequences:
    - 3 nickels
    - nickel, dime
    - dime, nickel
    - two dimes
  - draw state diagram:
    - inputs: N, D, reset
    - output: open chute
  - assumptions:
    - assume N and D asserted for one cycle
    - each state has a self loop for N = D = 0 (no coin)



Example: vending machine (cont'd)

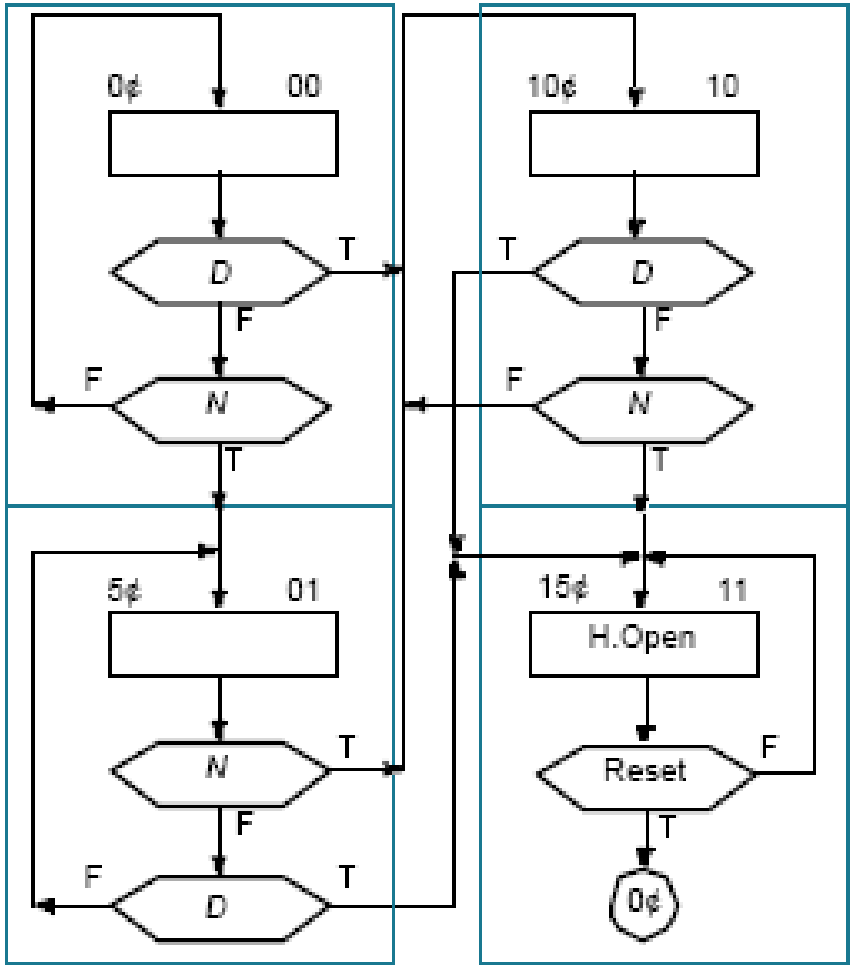
- Minimize number of states - reuse states whenever possible



present state	inputs D	N	next state	output open
0¢	0	0	0¢	0
	0	1	5¢	0
	1	0	10¢	0
	1	1	–	–
5¢	0	0	5¢	0
	0	1	10¢	0
	1	0	15¢	0
	1	1	–	–
10¢	0	0	10¢	0
	0	1	15¢	0
	1	0	15¢	0
	1	1	–	–
15¢	–	–	15¢	1

symbolic state table

ASM Chart for Vending Machine



ASMD Chart

- Form an ASMD (Algorithmic State Machine and datapath) chart by annotating each of its paths to indicate the concurrent register operations that occur in the associated datapath unit when the state of the controller makes a transition along the path
- Clarify a design of a sequential machine by separating the design of its datapath from the design of the controller
- ASMD chart maintains a clear relationship between a datapath and its controller
- Annotate path with concurrent register operations
- Outputs of the controller control the datapath

Example: Two-Stage Pipeline

